The Rules For The Manufacture, Use, Import, Export And Storage Of Hazardous Micro-Organisms/genetically Engineered organisms Or Cells, 1989

Union of India - Act

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India

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Rule

THE-RULES-FOR-THE-MANUFACTURE-USE-IMPORT-EXPORT-AND-STORAGE-OF-HAZARDOUS-MICRO-ORGANISMS-GENETICALLY-ENGINEERED-ORGANISMS-OR-CELLS-1989 of 1989

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The Rules For The Manufacture, Use, Import, Export And Storage Of Hazardous Micro-Organisms/genetically Engineered organisms Or Cells, 1989

Vide S.O. 1037(E), dated 5.12.1989, published in the Gazette of India, Ext., Point II, Section 3(i), dated 5.12.1989.

10.
/534
In exercise of the powers conferred by sections 6, 8 and 25 of the Environment (Protection) Act, 1986 (26 of 1986) and with a view to protecting the environment, nature and health, in connection with the application of gene-technology and micro-organisms, the Central Government hereby makes the following rules*, namely:
Additional Information6
*Draft Rules- " The Hazardous Micro Organisms and Genetically Modified Organisms (Manufacture, Use, Import and Storage) Rules, 1999, in supersession of the aforementioned Rules, published vide G.S.R. 98(E), dated 19.2.1999.
1.
Short title, extent and commencement.
(1)
These rules may be called The Rules For The Manufacture, Use, Import, Export And Storage Of Hazardous Micro-Organisms/genetically Engineered Organisms Or Cells, 1989.
(2)
These rules shall come into operation on the date to be notified for this purpose in the Official Gazette.
2.
Application.-
(1)
These rules are applicable to the manufacture, import and storage of micro-organisms and gene-technological products.
(2)
These rules shall also apply to genetically engineered organisms/micro-organisms and cells and correspondingly to any substances and products and food-stuffs, etc., of which such cells, organisms or tissues thereof form part.
(3)
These rules shall also apply to new gene-technologies apart from those referred to in clauses (ii) and (iv) of rule 3 and these rules shall apply to organisms/micro-organisms and cells generated by the utilisation of such other gene-technologies and to substances and products of which such organisms and cells form part.
(4)
These rules shall be applicable in the following specific cases
(a)
sale, offers for sale, storage for the purpose of sale, offers and any kind of handling over with or without a consideration;
(b)
exportation and importation of genetically engineered cells or organisms;
(c)
production, manufacturing, processing, storage, import, drawing off, packaging and repackaging of the genetically engineered products;
(d)
production, manufacture, etc., of drugs and pharmaceuticals and food-stuffs, distilleries and tanneries, etc., which make use of micro-organisms/genetically engineered micro-organisms one way or the other.
(5)
These rules shall be applicable to the whole of India.
3.
Definitions.-
In these rules, unless the context requires,
(i)
"Biotechnology" means the application of scientific and engineering principles to the processing of materials by biological agents to produce goods and services;
(ii)
"cell hybridisation" means the formation of live cells with new combinations of genetic material through the fusion of two or more cells by means of methods which do not occur naturally;
(iii)
["Expert Members:" Director General-Indian Council of Agricultural Research, Director General -Indian Council of

Medical Research, Director General-Council of Scientific and Industrial Research, Director GeneralHealth Services, Plant Protection Advisor, Directorate of Plant Protection, Quarantine and Storage, Chairman, Central Pollution Control Board or their representatives not below the rank of Joint Secretary and three outside experts in individual capacity.]
[Substituted by G.S.R. 1(E), 23rd December, 2010]
(iv)
"genetic engineering" means the technique by which heritable material which does not usually occur or will not occur naturally in the organism or cell concerned, generated outside the organism or the cell is inserted into said cell or organism. It shall also mean the formation of new combinations of genetic material by incorporation of a cell into a host cell, where they occur naturally (self-cloning as well as modification of an organism or in a cell by deletion and removal of parts of the heritable material;
(v)
"micro-organisms" shall include all the bacteria, viruses, fungi, mycoplasma, cell lines, algae, protozoans and nematodes indicated in the Schedule and those that have not been presently known to exist in the country.
Prior to substitution by by G.S.R. 1(E), 23rd December, 2010 on 23.12.2010 the clause (iii) read under;
(iii) "gene technology" means the application of the gene technique called genetic engineering, include self-cloning and deletion as well as cell hybridisation;
4.
Competent Authorities.-
(1)
Recombinant DNA Advisory Committee (RDAC).-This Committee shall review developments in Biotechnology at national and international levels and shall recommend suitable and appropriate safety regulations for India in recombinant research, use and applications from time to time. The Committee shall function in the Department of Biotechnology.
(2)
Review Committee on Genetic Manipulation (RCGM).-This Committee shall function in the Department of Biotechnology to monitor the safety related aspects in respect of on-going research projects and activities involving genetically engineered organisms/ hazardous micro-organisms. The Review Committee on Genetic Manipulation shall include representatives of (a) Department of Biotechnology (b) Indian Council of Medical Research (c) Indian Council of Agricultural Research (d) Council of Scientific and Industrial Research (e) other experts in their individual capacity. Review Committee on Genetic Manipulation may appoint sub-groups.
It shall bring out manuals of guidelines specifying procedure for regulatory process with respect to activities involving genetically engineered organisms in research, use and applications, including industry, with a view to ensure environmental safety. All on-going projects involving high risk category and controlled field experiments shall be reviewed to ensure that adequate precautions and containment conditions are followed as per the guidelines.
The Review Committee on Genetic Manipulation shall lay down procedures restricting or prohibiting production, sale, importation and use of such genetically engineered organisms or cells as are mentioned in the Schedule.
(3)
Institutional Biosafety Committee (IBSC).-This Committee shall be constituted by an occupier or any person, including research institutions, handling micro-organisms/genetically engineered organisms. The Committee shall comprise the Head of the Institution, scientists engaged in DNA work, a medical expert and a nominee of the Department of Biotechnology. The occupier or any person including research institutions handling micro-organisms/genetically engineered organisms shall prepare, with the assistance of the Institutional Biosafety Committee (IBSC), an up-to-date on-site emergency plan according to the manuals/ guidelines of the RCGM and make available copies to the District Level Committee/State Biotechnology Co-ordination Committee and the Genetic Engineering Approval Committee.
(4)
Genetic Engineering Approval Committee (GEAC).-This Committee shall function as a body under the Department of Environment, Forests and Wildlife for approval of activities involving large scale use of hazardous micro-organisms and recombinants in research and industrial production from the environmental angle. The Committee shall also be responsible for approval of proposals relating to release of genetically engineered organisms and products into the environment, including experimental field trials.
(5)

State Biotechnology Co-ordination Committee (SBCC).-There shall be a State Biotechnology Co-ordination Committee in the States wherever necessary. It shall have powers to inspect, investigate and take punitive action in case of violation of statutory provisions through the Nodal Department and the State Pollution Control Board /Directorate of Health/Medical Services. The Committee shall review periodically the safety and control measures in the various industries/ institutions handling genetically engineered organisms /hazardous micro-organisms.

(6)

District Level Committee (DLC).-There shall be a District Level Biotechnology Committee (DLC) in the districts wherever necessary under the District Collectors to monitor the safety regulations in installations engaged in the use of genetically modified organisms /hazardous micro-organisms and its applications in the environment.

The District Level Committee or any other persons authorised in this behalf shall visit the installation engaged in activity involving genetically engineered organisms, hazardous micro-organisms, formulate an information chart, find out hazards and risks associated with each of these installations a. . co-ordinate activities with a view to meeting any emergency. They shall al,,,) prepare an off-site emergency plan. The District Level Committee shall regularly submit its report to the State Biotechnology Co-ordination Committee and the Genetic Engineering Approval Committee.

5.

Classification of micro-organisms or genetically engineered product.-

(1)

For the purpose of these rules, micro-organisms or genetically engineered organisms, products or cells shall be dealt with under two major heads: animal pathogens and plant pests and these shall be classified in the manner specified in the Schedule.

(2)

If any of the micro-organisms/genetically engineered organism or cell falls within the limits of more than one risk class as specified in the Schedule, it shall be deemed to belong exclusively to the last in number of such classes.

6.

Micro-organisms laid down in the Schedule are divided into the following.

(i)

Bacterial Agents;

(ii)

Fungal Agents;

(iii)

Parasitic Agents;

(iv)

Viral, Rickettsial and Chlamydial Agents;

(v)

Special Category.

7.

Approval and prohibition, etc.-

(1)

No person shall import, export, transport, manufacture, process, use or sell any hazardous micro-organism or genetically engineered organism/ substance or cell except with the approval of the Genetic Engineering Approval Committee.

(2)

Use of pathogenic micro-organism or any genetically engineered organism or cell for the purpose of research shall only be allowed in laboratories or inside laboratory areas notified by the Ministry of Environment and Forests for this purpose under the Environment (Protection) Act, 1986.

(3)

The Genetic Engineering Approval Committee shall give directions to the occupier to determine or take measures concerning the discharge of micro-organisms/genetically engineered organisms or cells mentioned in the Schedule from the laboratories, hospitals and other areas including prohibition of such discharges and laying down measures to be taken to prevent such discharges.

(4)

Any person operating or using genetically engineered organisms/ micro-organisms mentioned in the Schedule for scale up or pilot operations shall have to obtain licence issued by the Genetic Engineering Approval Committee for any such activity. The possessor shall have to apply for licence in the prescribed profornia.

(5)

Certain experiments for the purpose of education within the field of gene technology or micro-organism may be carried outside the laboratories and laboratory areas mentioned in sub-rule (2) and will be looked after by the Institutional Biosafety Committee.

8.

Production.-

Production in which genetically engineered organisms or cells or micro-organisms are generated or used shall not be commenced except with the consent of Genetic Engineering Approval Committee with respect to the discharge of genetically engineered organisms or cells into the environment. This shall also apply to production taking place in connection with development, testing and experiments where such production, etc., is not subject to rule 7.

9.

Deliberate or unintentional release.-

(1)

Deliberate or unintentional release of genetically engineered organisms /hazardous micro-organisms or cells, including deliberate release for the purpose of experiment shall not be allowed.

Note.-Deliberate release shall mean any intentional transfer of genetically engineered organisms/ hazardous microorganisms or cells to the environment or nature, irrespective of the way in which it is done.

(2)

The Genetic Engineering Approval Committee may in special cases give approval of deliberate release.

10. Permission and approval for certain substances.

-Substances and products, which contain genetically engineered organisms or cells or micro-organisms shall not be produced, sold, imported or used except with the approval of the Genetic Engineering Approval Committee.

11.

Permission and approval for food-stuffs.

-Food-stuffs, ingredients in food-stuffs and additives, including processing aids, containing or consisting of genetically engineered organisms or cells, shall not be produced, sold, imported or used except with the approval of the Genetic Engineering Approval Committee.

12.

Guidelines.-

(1)

Any person who applies for approval under rules 8-11 shall, as determined by the Genetic Engineering Approval Committee submit information and make examinations or cause examinations to be made to elucidate the case, including examinations according to specific directions and at specific laboratories. He shall also make available an on-site emergency plan to the Genetic Engineering Approval Committee before obtaining the approval. If the authority itself makes an examination it self, it may order the applicant to defray the expenses incurred by it in doing so.

(2)

Any person to whom an approval has been granted under rules 8-11 above shall notify the Genetic Engineering Approval Committee of any change in or addition to information already submitted.

13.

Grant of approval.

(1)

In connection with the granting of approval under rules 8-11 above, terms and conditions shall be stipulated, including terms and conditions as to the control to be exercised by the applicant, supervisions, restriction on use, the layout of the enterprise and as to the submission of information to the State Biotechnology Co-ordination Committee or to the District Level Committee.

(2)

All approvals of the Genetic Engineering Approval Committee shall be for a specified period not exceeding four years at the first instance renewable for 2 years at a time. The Genetic Engineering Approval Committee shall have powers to revoke such approval in the following situations:

(a)

if there is any new information as to the harmful effects of the genetically engineered organisms or cells;

(b)

if the genetically engineered organisms or cells cause such damage to the environment, nature or health as could not be envisaged when the approval was given; or

(c)

non-compliance of any condition stipulated by the Genetic Engineering Approval Committee.

14.

Supervision.-

(1)

The Genetic Engineering Approval Committee may supervise the implementation of the terms and conditions laid down in connection with the approvals accorded by it.

(2)

The Genetic Engineering Approval Committee may carry out this supervision through the State Biotechnology Co-ordination Committee or the State Pollution Control Board/District Level Committee-or through any person authorised in this behalf.

15.

Penalties.-

(1)

If an order is not complied with, the District Level Committee or State Biotechnology Co-ordination Committee may take measures at the expense of the person who is responsible.

(2)

In case where immediate intervention is required in order to prevent any damage to -the environment, nature or health, the District Level Committee or State Biotechnology Co-ordination Committee may take the necessary steps without issuing any orders or notice. The expenses incurred for this purpose will be repayable by the person responsible for such damage.

(3)

The State Biotechnology Co-ordination Committee /District Level Committee may take samples for a more detailed examination of organisms and cells.

(4)

The State Biotechnology Co-ordination Committee/ District Level Committee shall be competent to ask for assistance from any other Government authority to carry out its instructions.

16.

Responsibility to notify interruption or accidents.-

(1)

Any person who under rules 7-11 is responsible for conditions or arrangements shall immediately notify the District Level Committee, State Biotechnology Co-ordination Committee and the State medical officer of any interruption of operations or accidents that may lead to discharges of genetically engineered organisms or cells which may be harmful to the environment, nature of health or involve any danger thereto.

(2)

Any notice given under sub-rule (1) above shall not lessen the duty of the person who is responsible to try effectively to minimise or prevent the effects of interruption of operations or accidents.

17.

Preparation of Off-site Emergency Plan by the DLC.

(1)

It shall be the duty of the DLC to prepare an off-site emergency plan detailing how emergencies relating to a possible major accident at a site will be dealt with and in preparing the plan, the DLC shall consult the occupier and such other person as it may deem necessary.

(2)

For the purpose of enabling the DLC to prepare the emergency plan required under sub-rule (1), the occupier shall provide the DLC with such information relating to the handling of hazardous micro-organisms/ genetically engineered organisms under his control as the DLC may require, including the nature, extent and likely off-site effects of a possible major accident and the DLC shall provide the occupier with any information from the off-site emergency plan which relates to his duties under rule 16.

18.

Inspection and information regarding finance.-

(1)

The State Biotechnology Co-ordination Committee or the Genetic Engineering Approval Committee/the DLC or any person with special knowledge duly authorised by the State Biotechnology Co-ordination Committee or the Genetic Engineering Approval Committee or the DLC where it is deemed necessary, at any time on due production of identity be admitted to public as well as to private premises and localities for the purpose of carrying out supervision.

(2)

Any person who is responsible for activities subject to rules 7-11 above shall at the request of District Level Committee or State Biotechnology Co-ordination Committee or the Genetic Engineering Approval Committee submit all such information, including information relating to financial conditions and accounts, as is essential to the authority's administration under these rules. He shall also follow supervision or inspection by the authorities or persons indicating in sub-rule (1).

(3)

The Genetic Engineering Approval Committee may fix fees to cover, in whole or in part, the expenses incurred by the authorities in connection with approvals, examination, supervision and control.

19.

Appeal.-

(1)

Any person aggrieved by a decision made by the Genetic Engineering Approval Committee/State Biotechnology Co-ordination Committee in pursuance of these rules may within thirty days from the date on which the decision is communicated to him, prefer an appeal to such authority as may be appointed by the Ministry of Environment and Forests provided that the Appellate Authority may entertain the appeal after the expiry of the said period of thirty days if such authority is satisfied that the appellant was prevented by sufficient cause from filing the appeal in time.

20.

Exemption.

-The Ministry of Environment and Forests shall, wherever necessary, exempt an occupier handling a particular micro-organism/ genetically engineered organism from rules 7-11.

Schedule

A. ANIMAL AND HUMAN PATHOGENS

BACTERIAL

Risk Group II

Acinetobacter calcoacetieus

Actinobacillus-all species except A mallei, which is in Risk Group III

Aeromonoas hydrophila

Arizona hinshawii-all serotypes

Bacillus anthracis

Bordetella all species

Borrelia recurrentis.B.Vincenti

Campylobacter fetus

Camphylobacter jejuni, Chalamydia psittaci

Cheamydia trachomatics

Clostridium chauvoei, Cl.difficile Cl/fallax. Cl haemolyticum Q.histolyticum, CI novyi

(CI,Pefringes) Cl.speticum, Cl.sordelli

Corynebacterium diptheriae, C.equi, C. haemolyticum, C.Pseudotuberculosis, C.pyogenes,

C.renale

Diplococcus

(Streptococcus)
pneumoniae
Edwardsiila tarda
Erysipelothix insidiosa
Escherichia Coli-all enteropathogenic serotypes, enterotoxigenic
Haemophilus ducrevi,
H.influenzae, H. pneumoniae
Herellea vaginicola
Klebsiella-all species and all serotypes
Legionlla pneumophila
Letionella
Leptospira interrogans-all serotypes reported in India
Listeria, all species
Mima polymorpha
Moraxella-all species
Mycobacteria-all species including Mycobacterium avium
M.Bovis M.tuberculosis, M.Leprae
Mycoplasma-all species except M.Mycoides and M.angalactiae
Meosseroc gonorrhoea,N. Leprae
Mycoplasma-all species except M.Mycoides and M.angalactiae
Neisseric gonorrhoea,N.
meningitis
Pasteurella-all species except those listed in Risk Group III
Salmonella-all species and all setotypes
Shigella-all species and all serotypes
Sphaerophorgs necrophorus
Staphylococcus aureus
Streptobacillus moniliformis
Streptococcus pneumoniae
Streptococcus pyogenes.S.equi
Streptomyces madurae,s.pelleteri, s.somaliensis
Treponema carateum, T.pallidam and T.pettenue
Vibrio foetus V.comma including biotype EI Top and
V.
parahemolyticus
Vibrio cholerae
Risk Group III:
Actinobacillus mallei
Bartonella-all species
Brucella-all species
Clostridium botulium Cl.tetani
Francisella tularensis
Mycobacterium avium,. M.bovis, M.tuberculosis,m.leprae
Pasteurella multocida type B("buffalo" and other foreign virulent strains)
Pseudomonas pseudomallai
Yersinia pestis
FUNGAL
Risk Group II
· Actinomycetes
(including Nocardia SP, Actinomyces species and Arachina propinica)
· Aspergillus fumigatus

· Blastomyces dermatitis
· Cryptococcus neoformans C. fersiminosos
· Epidermophyton madurella, microsporon
· Paracoccidiodes brasiliensis
· Sporothrix
· Trichoderma
· Trichophyton

Risk Group III

Coccidioides immitis

Histoplasma capulatum

Histoplasma capsulatum var duboiss

PARASITIC

Risk Group II

· Entahoeba histolytica
· Leishmania species
· Naegleria gruberia
· Plasmodium theilera, P. babesia, P. falcoparum
· Plasmodium babesia
· Schistosoma
· Toxoplasma gondii
· Toxocana canis
· Trichinella spiralis
· Trichomanas
· Trypanosoma cruzi

Risk Group III

· Schisistosoma mansoni

VIRAL RICKETTSIAL AND
CHALMYDIAL

Risk Group II

· Adenoviruses

- Human all types

· Avian loukosis
· Cache Valley virus
· CELO

(avian adenovirus)

· Coxsackie A and B viruses
· Corona viruses
· Cytomegalo viruses
· Dengue virus, when used for transmission experiments
· Echo viruses - all types
· Encephalomyocarditis virus (EMC)
· Flanders virus
· Hart Part virus
· Hepatitis

- associated antigen material - hepatitis A and B viruses, non A and non B, HDV

· Herpes viruses - except herpesviruses simiae (monkey B virus) which is in Risk Group IV.
· Infectious Bovine Rhinotraechitis virus (IBR)
· Infectious Bursal diseases of poultry and Infectious Bronchitus
· Infectious Laryngotraechitis (ILT)
· Influenza virus - all types, except A PR 834 which is in Risk Group I
· Langat virus Leucosis Complex

· Lymphogranuloma venereum agent
· Marek's Disease virus
· Measles virus
· Mumps virus
· Newcastle disease virus (other than licenced strain for vaccine use)
· Parainfluenza viruses - all type except parainfluenza virus 3, SF4 strain, which is in Risk Group I.
· Polio viruses - all types, wild and attenuated
· Poxviruses

- all types except Alastrim, monkey pox, sheep pox and white pox, which depending on experiments are in Risk Group III or IV.

· Rabies virus - all strains except rabies stret virus, which should be classified in Risk Group III when inoculated into cornivores

· Reoviruses

- all types

· Respiratory syncytial virus
· Rhinoviruses

- all types

· Rinderpest

(other than vaccine strain in use)

· Rubella virus
· Stimian viruses - all types except herpeavirus simlae (Monkey Virus) which is in Risk Group IV.
· Simian virus 40 -
· Ad 7 SV 40 (defective)
· Sindbis virus
· Tensaw virus
· Turlock virus
· Vaccinia virus
· Varicella virus
· Vole rickettsia
· Yellow fever virus, 17D vaccine strain

Risk Group III

African House Sickness (attenuated strain except animal passage)

Alastrim,

monkey pox and whitepox, when used in vitro

Arboviruses -

All strains except those in Risk Group II and IV.

Blue tongue virus (only serotypes reported in India)

Ebola fever virus

Feline Leukemia Epstein-Barr virus

Feline sarcoma

Foot and Mouth Disease virus (all serotypes and sutbypes)

Gibbon Ape Lymphosarcoma

Herpesvirus ateles

Herpesvirus saimiri

Herpes simplex 2

HIV-I & HIV-2 and strains of SIV

Infectious Equine Anaemia

Lymphocytic choriomeningitis virus (LCM)

Monkey pox,

when used in vitro

Nen-defective Adeno-2 SV-40 hybrids

Psittacosis-ornithosis-trachoma group of agents
Pseudorabies virus
Rabies street virus, when used inoculations of carnivores
Rickettsia-all species except Vole rickettsia and Coxiell burnetti when used for vector transmission or animal inoculation experiments
Sheep pox
(field strain)
Swine Fever virus
Vesicular stomatitis virus
Woolly monkey Fibrosarcoma
Yaba pox virus
Risk Group IV
Alastrim,
monkeypox, whitepox, when used for transmission or animal inoculation experiments
Hemorrhagic fever agents, including Crimean hemorrhagic fever (congo)
Korean hemorrhagic fever and others as yet undefined
Herpesvirus simlae (monkey B virus)
Tick-borne encephalitis virus complex, including - Russian
Spring Summer Encephalitis, Kyasanur Forest Diseast, omsk hemorrhagic fever and Central European encephalitis viruses
SPECIAL CATEGORY
BACTERIAL
Contagious Equine Metritis (H. equigenitalis)
Pestis petit de ruminantium
VIRAL RICKETTSIAL AND
CHLAMYDIAL
African Horse Sickness virus (serotypes not reported in India and challenge strains)
African Swine Fever
Bat rabies virus
Blue tongue virus (serotypes not reported in India)
Exoitic FMD virus types and sub-types
Junin and Machupo viruses
Lassa virus
Marburg virus
Murrey valley encephalitis virus
Rift Valley Fever virus
Smallpox virus
- Archieval storage and propagation Swine Vesicular Disease
Veneseulan equine encephalitis virus - epidemic strains
Western Equine encephalitis virus Yellow fever virus - Wild strain
Other Arboviruses causing epizootics and so far not recorded in India
B. PLANT PESTS
Any living stage (including
active and dormant forms) of insects, mites nematodes, slugs, snails, bacteria, fungi, protozoa, other parsitic plants or reproductive parts thereof: viruses;
or any organisms similar to or allied with any of the foregoing; or any
infectious agents or substances, which can directly or indirectly injure or
cause disease or damage in or to any plants or parts thereof, or any processed, manufactured, or other products of plants are considered plant pests.
Organisms belonging to all
lower Taxa contained within the group listed are also included.

1. Viruses:
All viroids
All bacterial, fungal, algal, plant, insect and nematode viruses; special care should be take for:
i.
i.
Geminiviruses,
ii.
ii.
Caulimoviruses,
iii.
iii.
Nuclear Polyhedrosis viruses,
iv.
iv.
Granulosis viruses, and
v.
v.
Cytoplasmic polyhedrosis viruses.
2. Bacteria:
Family Pseudomonadaceae
Genus Pseudomonas
Genus Xanthomonas
Genus Azotobacter
Family Rhizobiaceae
Genus Rhizobium/Azorhizobium
Genus Bradyrhizobium
Genus Agrobacterium
Genus Phyllobacterium
Genus Erwinia
Genus Enterobacter
Genus Klebzieller
Family Spirollacea
Genus Azospirillum
Genus Acquspirillum
Genus Oceonospirillum
Family Streptomycetaceae
Genue Streptomyces
Genue Nocardia
Family Actinomycetaceae
Genue Actinomyces
Coryneform Group
Genus Clavibacter
Genus Arthrobacter
Genus Curtobacterium
Genus Bdellovibro
Family Rickettsiaceae
Rickettsial-like
organisms associated with insect diseases
Gram-negative phloem-limited
bacteria associated with plant diseases
Gram-negative xylem-limited

bacteria associated with plant diseases
Cyanobacteria - All members of
blue-green algae
Mollicutes
Family Spiroplasmataceae
Mycoplama-like organisma
associated with plant diseases
Mycoplasma-like organisms
associated with insect diseases
Algae
Family Chlorophyceae
Family Euglenophyceae
Family Pyrophyceae
Family Chrysophyceae
Family Phaephyceae
Family Rhodophyceae
Fungi
Family Plasmodiophoraceae
Family Chytridiaceae
Family Olpidiopsidaceae
Family Synchytriaceae
Family Catenariaceae
Family Coelomomycetaceae
Family Saprolegniaceae
Family Zoopagaceae
Family Albuginaceae
Family Peronosporaceae
Family Pythiaceae
Family Mucoraceae
Family Choanephoraceae
Family Mortiercllaceae
Family Endogonaceae
Family Syncephalastraceae
Family Dimargaritaceae
Family Kickxellaceae
Family Saksenaeaceae
Family Entomophthoraceae
Family Ecerinaceae
Family Taphrinaceae
Family Endomycetaceae
Family Saceharomycetaceae
Family Eurotiaceae
Family Gymnoascaceae
Family Aseophaeriaceae
Family Onygenaceae
Family Microascaceae
Family Protomycetaceae
Family Elsinoeaceae
Family Myriangiaceae
Family Dothidiaceae
Family Chaetothyriaceae

Family Parmulariaceae
Family Phillipsiellaceae
Family Hysteriaceae
Family Pleosporaceae
Family Melamomataceae
Family Ophiostomataceae
Family Aseosphaeriaceae
Family Erysiphaceae
Family Meliolaceae
Family Xylariaceae
Family Diaporthaceae
Family Hypoereaceae
Family Clavicipataceae
Family Phacidiaceae
Family Ascocorticiaceae
Family Hemiphacidiaceae
Family Dermataceae
Family Sclerotiniaceae
Family Cyttariaceae
Family Helosiaceae
Family Sarcostomataceae
Family Sarcoscyphaceae
Family Auricolariaceae
Family Ceratobasidiaceae
Family Corticiaceae
Family Hymenochaetaceae
Family Echinodintiaceae
Family Eistuliniaceae
Family Clavariaceae
Family Polyporaceae
Family Tricholomattaceae
Family Ustilaginaceae
Family Sporobolomycetaceae
Family Uredinaceae
Family Agaricaceae
Family Graphiolaceae
Family Pucciniaceae
Family Melampsoraceae
Family Gandodermataceae
Family Labonlbeniaceae
Family Sphaeropsidaceae
Family Melabconiaceae
Family Tuberculariaceae
Family Dematiaceae
Family Moniliaceae
Family Aganomucetaceae
Parasitic Weeds
Family Balanophoraceae-parasitic species
Family Cuscutaceae-parasitic species
Family Ttydonoraceae-parasitic species
Family Lauraceae-parasitic species Genus Cassytha

Family Lennoaceae-parasitic species
Family Loranthaceae-parasitic species
Family Myzodendraceae-parasitic species
Family Olacaceae-parasitic species
Family Orobanchaceae-parasitic species
Family Rafflesiaceae-parasitic species
Family Santalaceae-parasitic species
Family Scrophulariaceae-parasitic species
Protozoa
Genus Phytomonas
And all
protozoa associated with insect diseases.
Nematodes
Family Anguinidae
Family Belonolaimidae
Fmaily Caloosiidae
Family Criconematidae
Family Dolichodoridae
Family Fergusobiidae
Family Hemicycliophoridae
Family Heteroderidae
Family Hoplolaimidae
Family Meloidogynidae
Family Neotylenchidac
Family Nothotylenchidae
Family Paratylenchidae
Family Pratylenchidae
Family Tylenchidae
Family Tylenchulidae
Family Aphelenchoidiae
Family Longidoridae
Family Trichodoridae
Mollusca
Superfamily Planorbacea
Superfamily Achatinacea
Superfamily Arionacea
Superfamily Limacacea
Superfamily Helicacea
Superfamily Veronicellacea
Arthropoda
Superfamily Ascoidea
Superfamily Dermanyssoidea
Superfamily Erjophyoidea
Superfamily Tetranychoidea
Superfamily Eupodoidea
Superfamily Tydeoidea
Superfamily Erythraenoidea
Superfamily Trombidioidea
Superfamily Hydryphantoidea
Superfamily Tarasonemoidea
Superfamily Pyemotoidea

Superfamily Hemisarcoptoidea
Superfamily Acaroidea
Order Polydesmida
Family Sminthoridae
Family Forfieulidae
Order Isoptera
Order Thysanoptera
Family Acrididea
Family Gryllidae
Family Gryllacrididae
Faily Gryllotalpidae
Family Phasmatidae
Family Ronaleidae
Family Tettigoniidae
Family Tetragidae
Family Thaumastocoridae
Superfamily Piesmatoidea
Superfamily Lygacoidea
Superfamily Idiostoloidea
Superfamily Careoidea
Superfamily Pentatomoidea
Superfamily Pyrrhocoroidea
Superfamily Tingoidea
Superfamily Miroidea
Order Homoptera
Family Anobiidae
Family Apionidae
Family Anthribidae
Family Bostrichidae
Family Brentidae
Family Bruchidae
Family Buprestodae
Family Byturidae
Family Cantharidae
Family Carabidae
Family Ceambycidae
Family Chrysomelidae
Family Coecinellidae
Family Curculionidae
Family Dermestidae
Family Elalteridae
Family Hydrophilidae
Family Lyctidae
Family Meloidae
Family Mordellidae
Family Platypodidae
Family Scarabaeldae
Family Scolytidae
Family Selbytidae
Order Lepidoptera
Family Agromyzidae

Family Anthomiidae
Family Cecidomiidae
Family Chioropidae
Family Ephydridae
Family Lonchaeidae
Family Muscidae
Family Otitidae
Family Syrphidae
Family Tephritidae
Family Tipulidae
Family Apidae
Family Caphidae
Family Chalcidae
Family Cynipidae
Family Eurytomidae
Family Formicidae
Family Psilidae
Family Sircidae
Family Tenthredinidae
Family Torymidae
Family Xyloiopidae and
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